Protocol for using BEAT.m in MATLAB

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Table of contents

1 Introduction

2-19 Using the script for initial data processing

20-21 Using the script to review and modify processed data

Introduction

This script was designed to process ECG data from animal-borne tags: detecting heartbeats (R waves) semi-automatically, reviewing and correcting them manually, and producing a table of beat times, RR intervals and instantaneous heart rates. It was built so that data are easy both to process and to review and edit afterwards, whether by the original processor or by someone else.

The script is free to use but runs in MATLAB, which requires a license, plus MATLAB's Signal Processing Toolbox. It was tested in MATLAB R2026a (Update 5). At a minimum, the script needs a plain-text data file (e.g. .txt or .csv) containing a column of ECG values, plus either a time column (in seconds) or a known sampling rate. Depth and acceleration can also be included and are displayed alongside the ECG.

In brief, the script: (1) imports the data; (2) crops it to the period the tag was on the animal; (3) filters the ECG; (4) automatically detects candidate heartbeats; and (5) opens an interactive window to review and edit them, saving the results and the settings used.

This protocol has two parts. Part 1 covers the initial processing: setting up the folder structure, running the script, and checking the outputs. Part 2 covers reviewing and editing processed data, either your own or data processed by others.

The script has been tested on deployments from several species, but there is always room for improvement. **If you have any questions or would like to discuss future updates, please contact me at ncnemeth@hawaii.edu.**

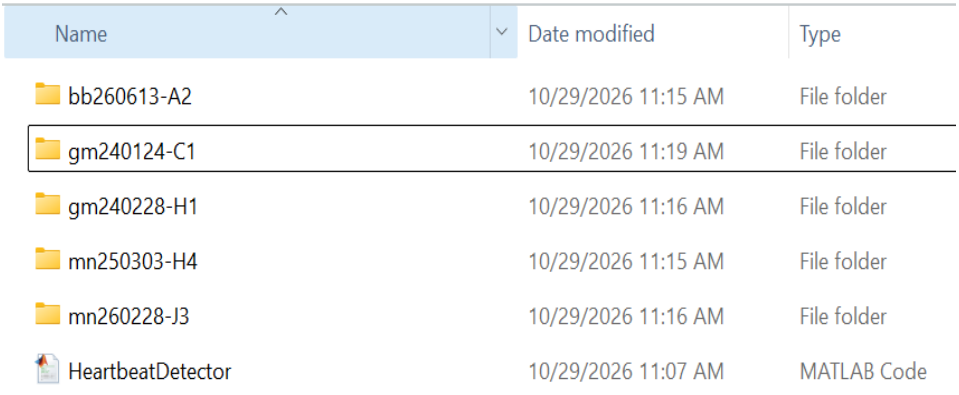
This script and protocol will be soon available on Github, but for now please contact me for how to cite the script.

This protocol was written for Version 4.0 of the script.

Using the script for initial data processing

File structure

Before opening the script for the first time, we need to make sure that your file structure is set up in the way that the script expects. You should have an uppermost level folder that contains all of your deployment folders and the BEAT script. In my case, I named this folder HeartRate, but anything will work. So inside of HeartRate (in my case) I would have one folder for each deployment, which looks something like this.



Inside of each deployment folder, at a minimum, needs a folder named "**Data**", with a subfolder "**Raw**", and inside of the "**Raw**" folder is the text file containing your raw data to be processed. The script will make a "**Processed**" folder automatically.

You can name the text file whatever you want. I name it the same as the deployment for consistency.

Note the uppermost level folder, "HeartRate" in our case, can live wherever you would like

See example folder tree on next page.

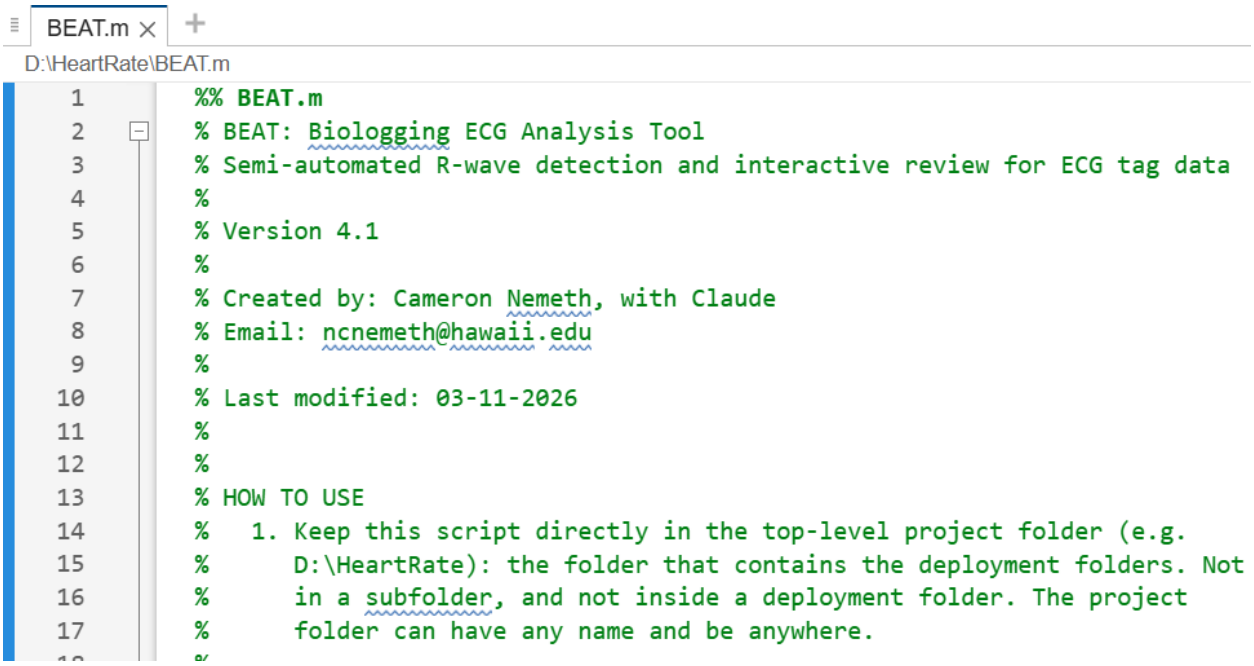
The path to your raw data should look like this:

```
HeartRate/
│
├── BEAT.m
├── BEAT_Protocol.pdf
│
├── bb260613-A2/
│   └── Data/
│       └── Raw/
│           └── bb260613-A2.txt
│
├── gm240124-C1/
│   └── Data/
│       └── Raw/
│           └── gm240124-C1.txt
│
└── mn250303-H4/
    └── Data/
        └── Raw/
            └── mn250303-H4.txt
```

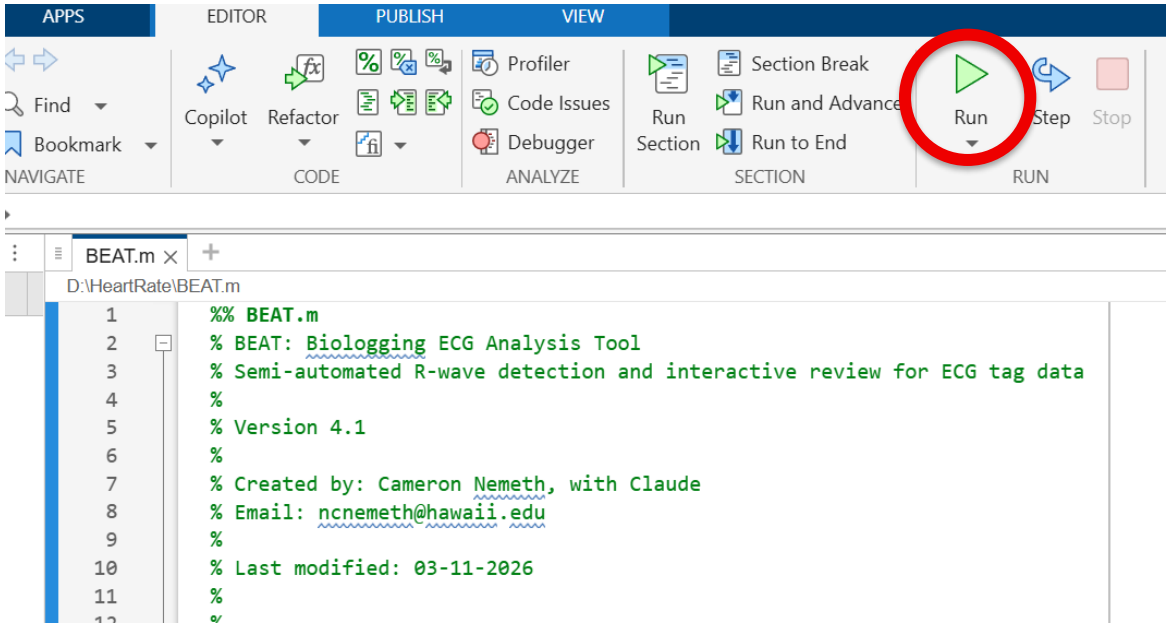
Opening the script for the first time

Once your file structure is set, you can open the BEAT.m script

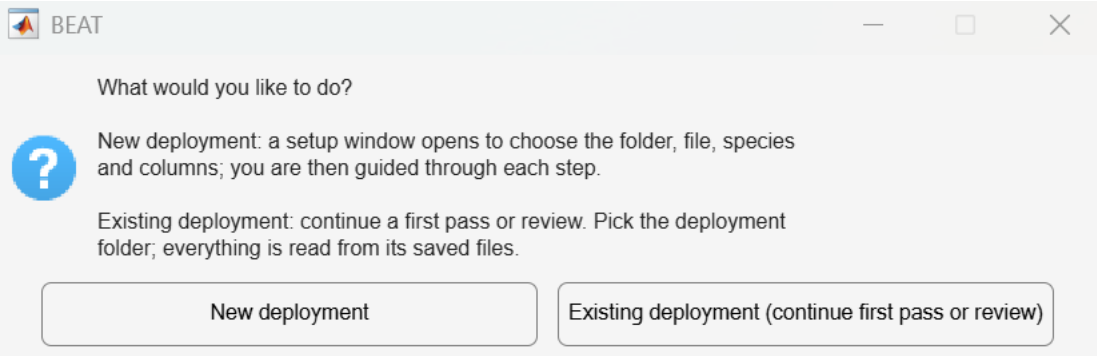
You should see an introduction at the top, which looks like this



Click "**Run**" along the top banner

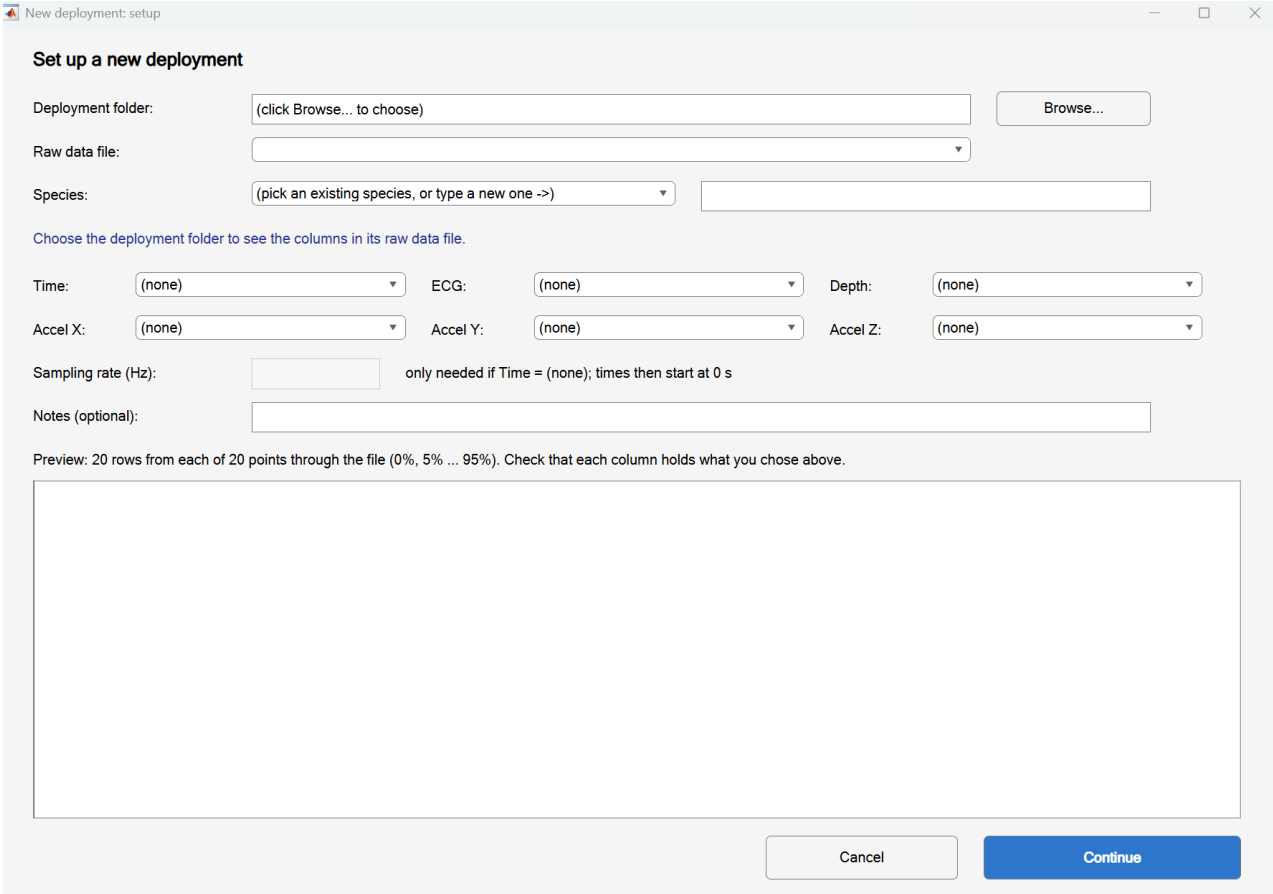


This popup box will appear. If you are working on a brand-new deployment, click "**New deployment"**. If you are continuing a previous deployment or reviewing past work, select "**Existing deployment**".

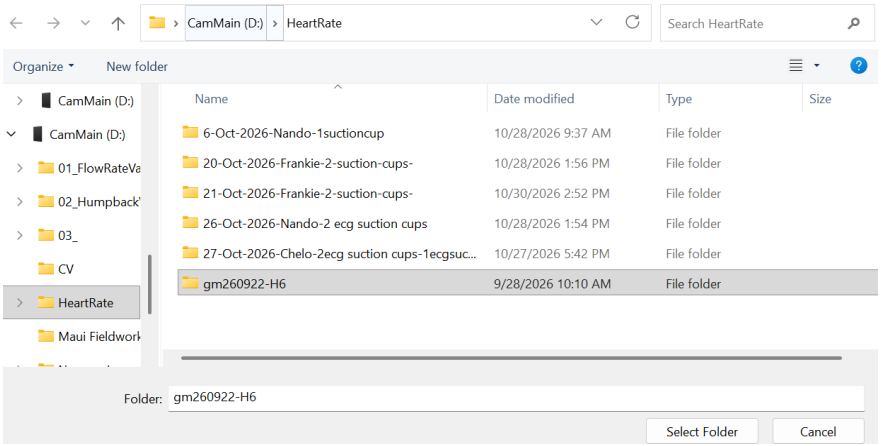


We will continue with an example of a new deployment for now.

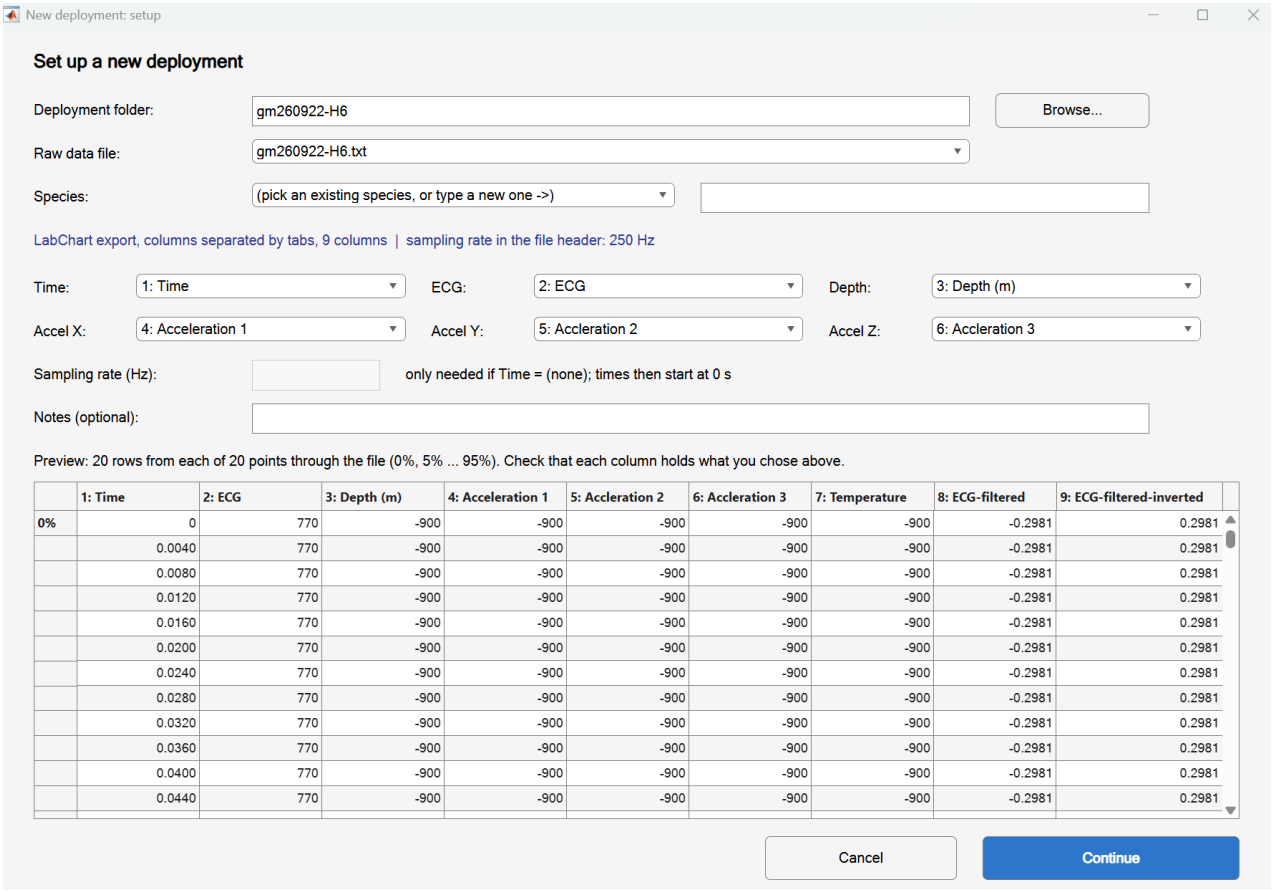
Another popup box will appear, asking you to select details regarding the deployment



Click “**Browse**” and select the deployment that you wish to work on.



After selecting the deployment, various fields will be automatically populated.



	1: Time	2: ECG	3: Depth (m)	4: Acceleration 1	5: Accleration 2	6: Accleration 3	7: Temperature	8: ECG-filtered	9: ECG-filtered-inverted
0%	0	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0040	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0080	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0120	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0160	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0200	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0240	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0280	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0320	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0360	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0400	770	-900	-900	-900	-900	-900	-0.2981	0.2981
	0.0440	770	-900	-900	-900	-900	-900	-0.2981	0.2981

Fill in any remaining sections that are not filled in.

“Species” has a drop-down list of previously used species; picking from it keeps names consistent across deployments. If the species is new, type its name in the box to the right.

The script makes its best guess at which column holds time, ECG, depth and acceleration, based on the column names. Check each choice against the preview table below: it shows 20 rows from each of 20 points through the file (0%, 5% … 95%).

To change a column, pick it from its drop-down list. If a channel was not recorded (e.g. no acceleration), choose “(none)”. ECG is required. If the file has no column names, columns are listed by number (“Column 1”, “Column 2”, …); use the preview to tell them apart.

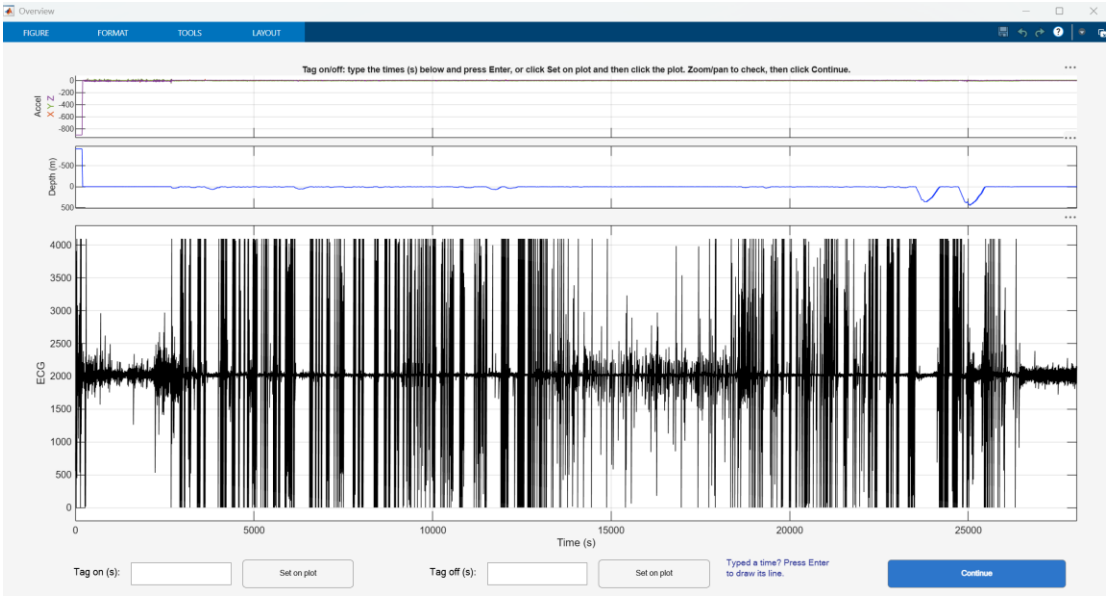
The sampling rate box is only needed if the file has no time column (Time = “(none)”).

Click “**Continue**” once all applicable sections are filled in

This step will read in the information from the raw text file, this may take a few seconds.
The script saves a .mat file of just the columns used by the rest of the script. Future runs load the .mat file instead to speed up loading times in the case of large datasets.

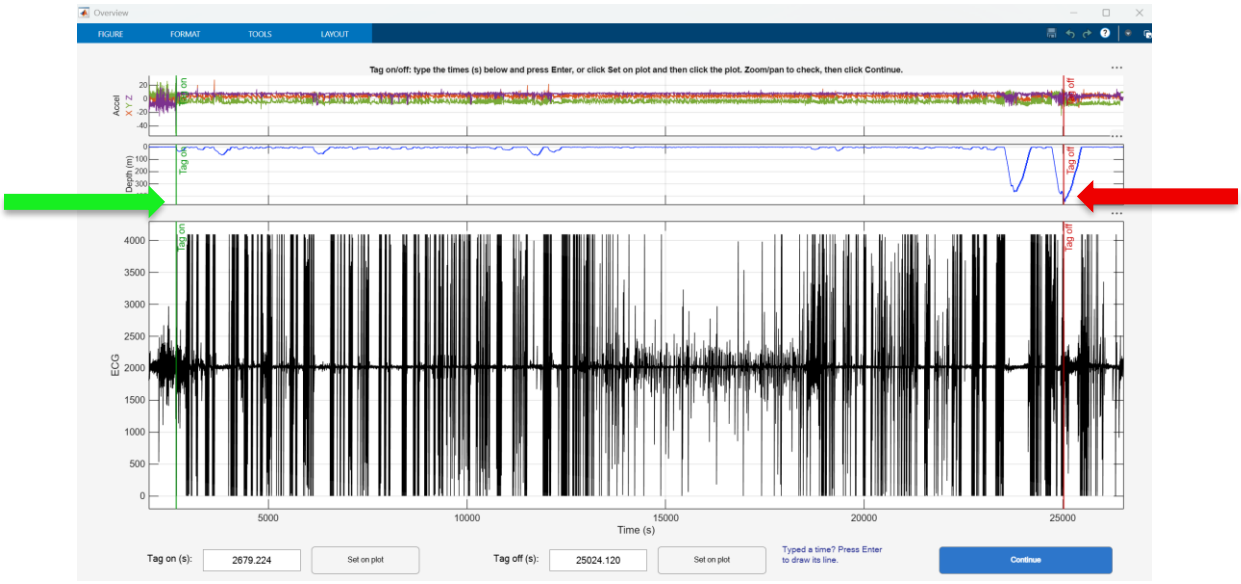
Cropping the data

Another window will appear asking you to select times when the tag went onto the animal and when it came off. Note: the ECG data will be shown along with depth and/or acceleration, if your raw data included those data streams. These will remain throughout

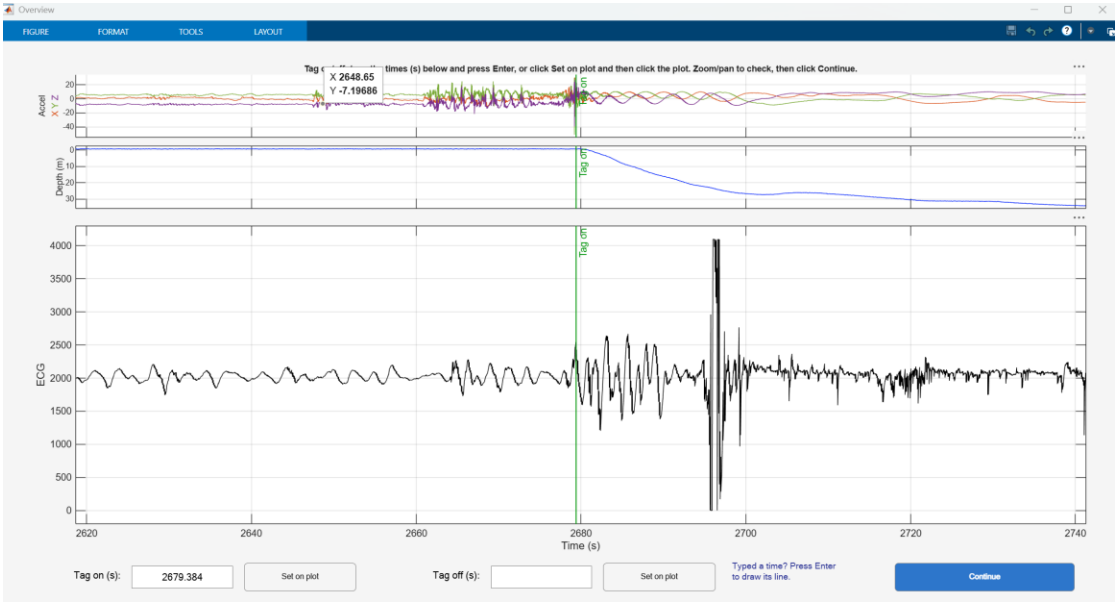


You have two options:

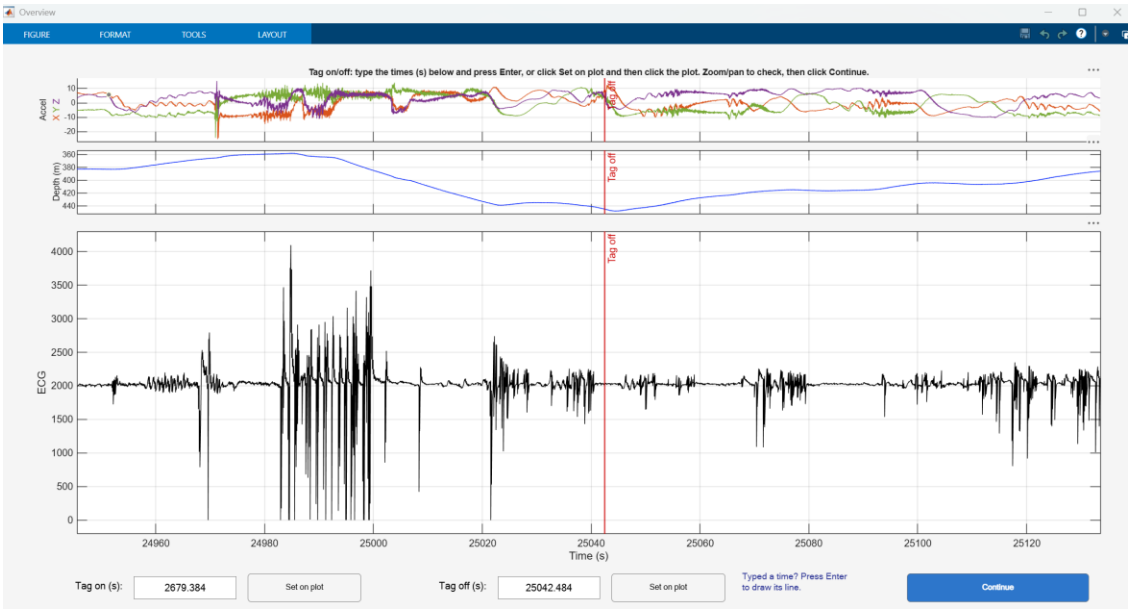
1. If you know the times, you may enter them into the boxes at the bottom and press enter after each one. You should see a green line with “Tag On” and red line with “Tag Off” appear on the plots.



2. If you wish to use other data streams (e.g., depth and/or acceleration) to determine these times, zoom into the data by scrolling and click and drag left to right to move around in the window. Find the moment the tag went on and THEN click "**Set on plot**". The moment you click "**Set on plot**" you lose the ability to move around the window.



Repeat for "**Tag off**"

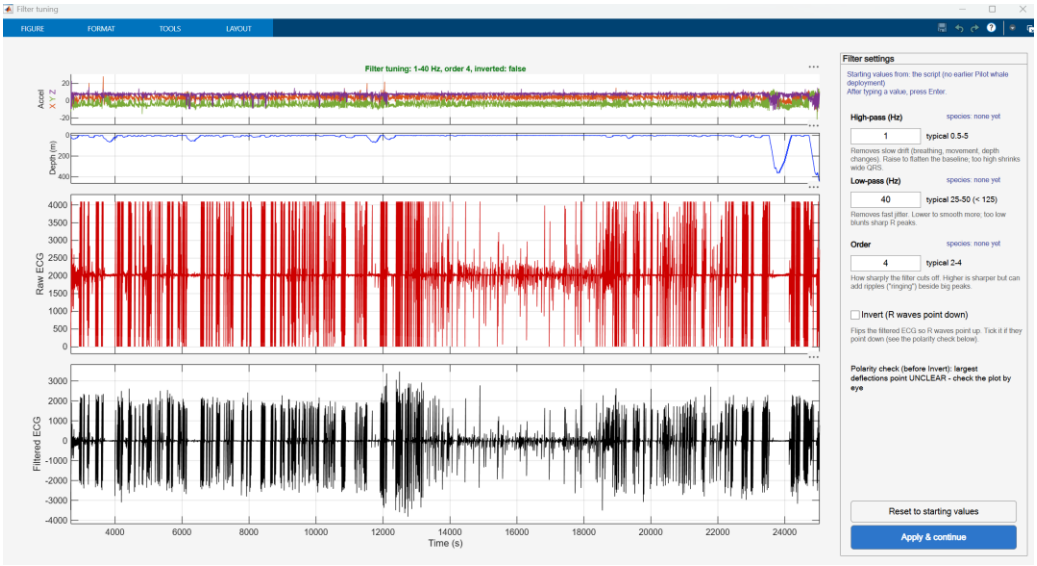


Click "**Continue**"

Note: While at this stage you may not be 100% certain about these times, it is important to do your best, as there are diagnostic outputs at the end that incorporate the deployment duration. It is easy to return and adjust these times later. The data will be cropped to your selection.

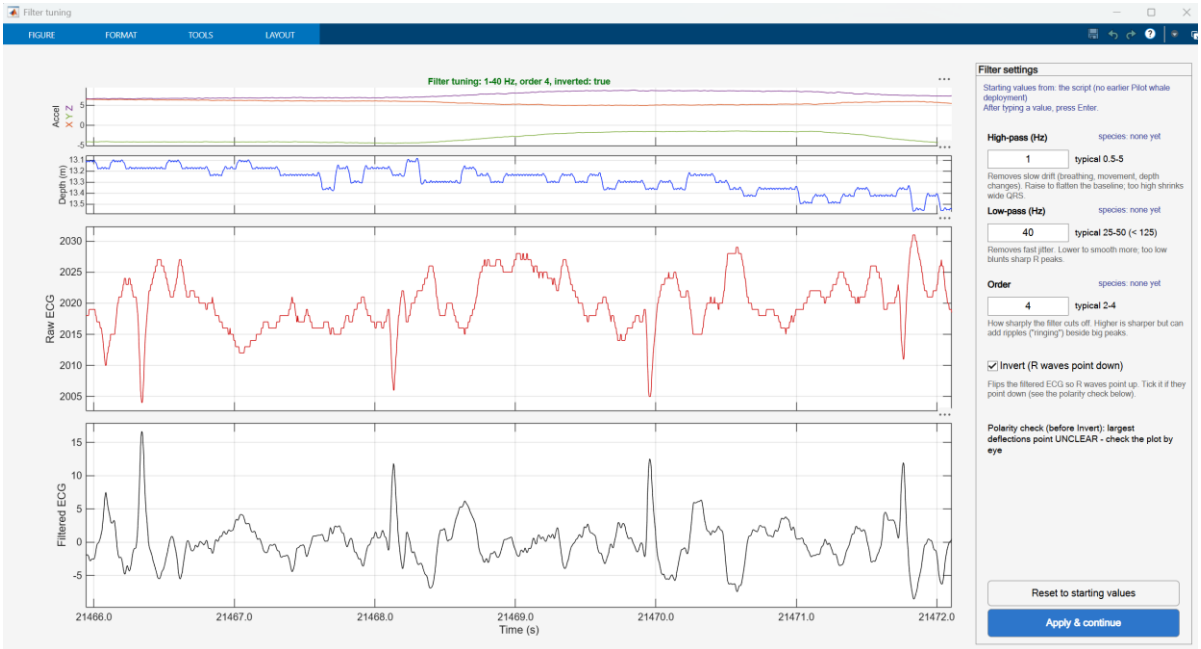
Filtering the data

A window will appear that looks like this.



Raw ECG signal (red), filtered signal (black), and depth and acceleration, if applicable.

On the right-hand side are options for a bandpass filter. These settings will be automatically set from the most recently processed deployment of the same species; otherwise from the default values in the script. The blue text at the top of the panel shows which was used. Each setting also shows its typical range and, in blue, the range used for this species in other deployments. Zoom into various parts of the data and change the settings as desired. **Remember to press enter after entering in new values** and the filtered data should be updated to reflect your changes.

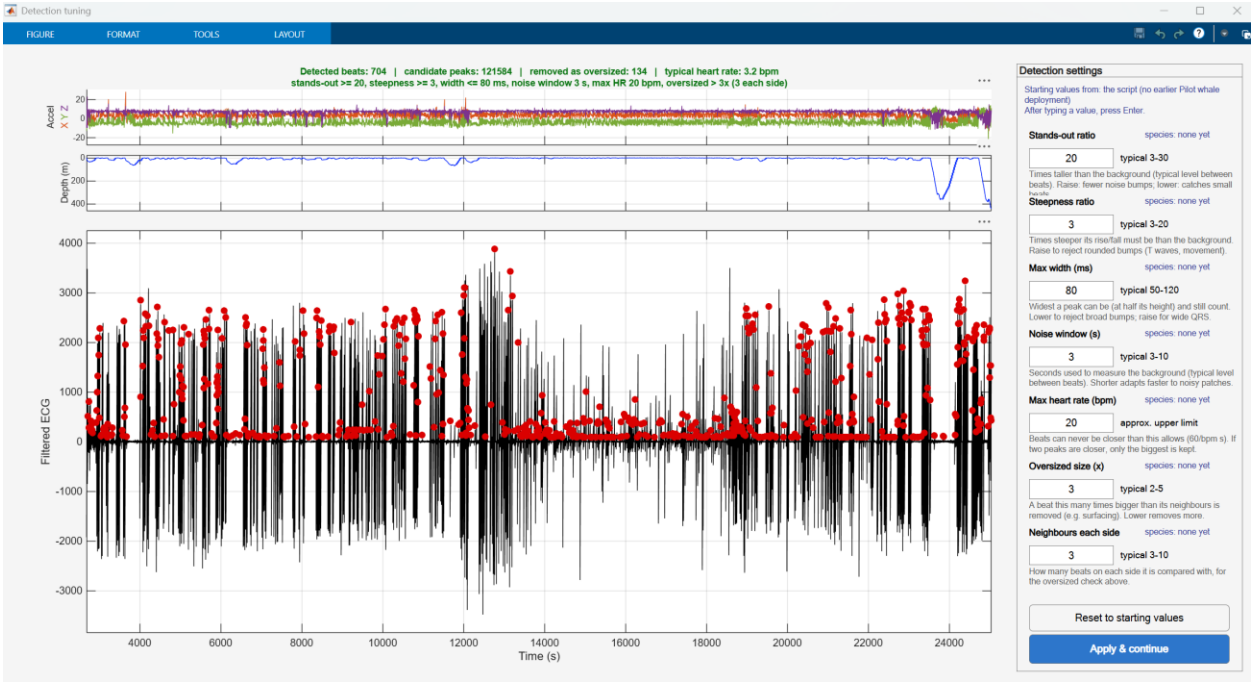


Click "**Apply and continue**" when finished.

Automatic R-wave detection

A window will appear showing the script's attempt to detect R waves automatically. Each red circle marks a peak the script has detected as an R wave. Some may be missed or wrong; the goal here is to get as many correct as possible, and any remaining errors are fixed in the review.

Note: This can be difficult, particularly with noisy data. My recommendation is to find a clean section and attempt to get as many peaks as possible. You will never get 100%, but the next step makes it very easy to clean these data.

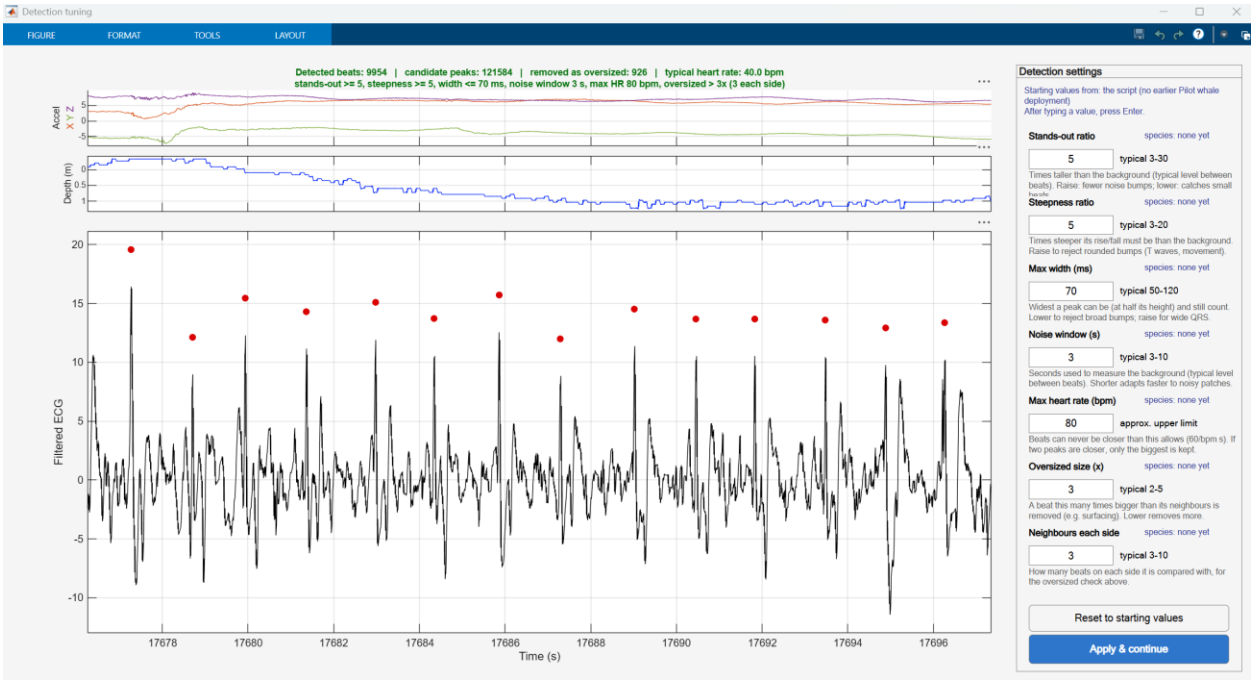


On the right-hand side are options for automated R-wave detection. These settings will be automatically set from the most recently processed deployment of the same species; otherwise from the default values in the script. The blue text at the top of the panel shows which was used. Each setting also shows its typical range and, in blue, the range used for this species in other deployments.

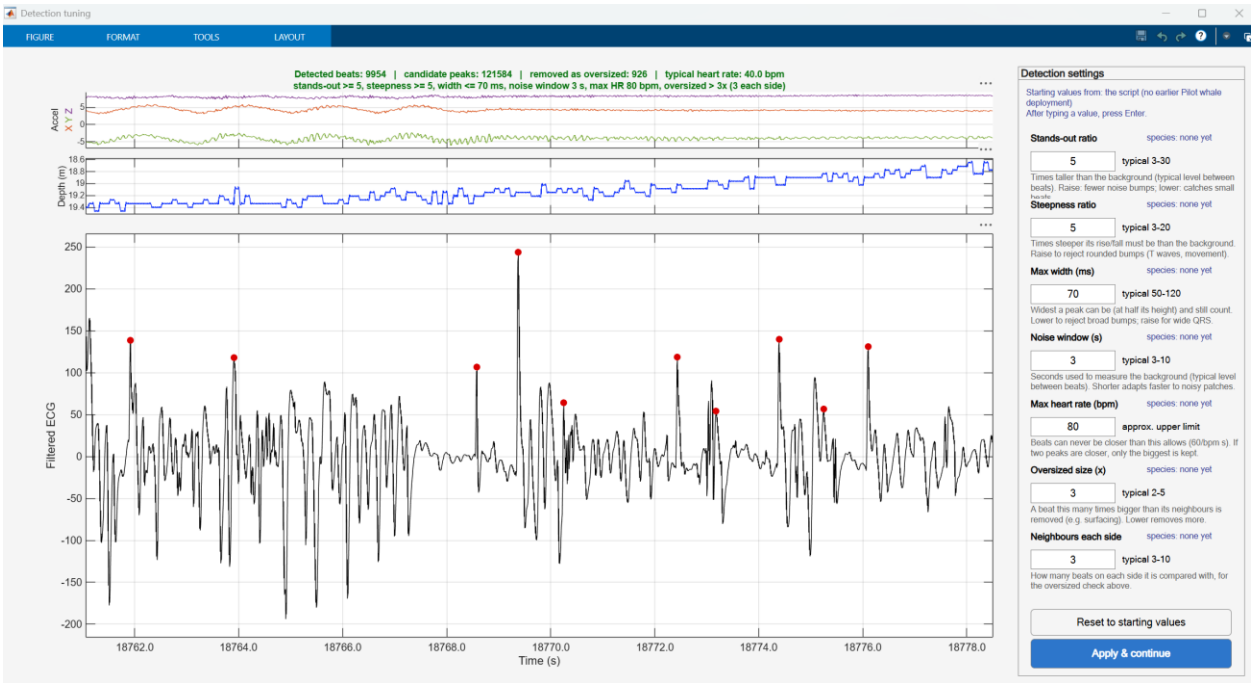
Each setting also has a short description beneath its entry box.

Zoom into various parts of the data and change the settings as desired. **Remember to press enter after entering in new values** and the filtered data should be updated to reflect your changes.

Certain settings may work really well on certain sections



While the same settings on a different section may not look as good.



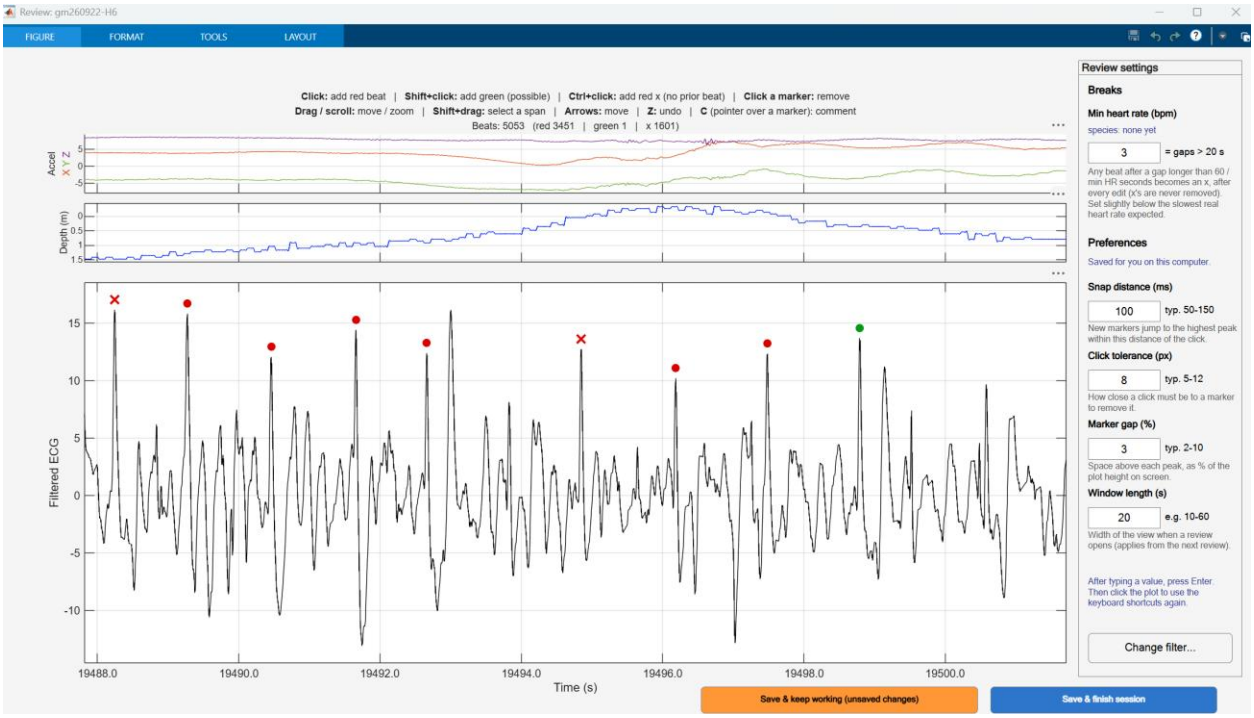
However, **if you can get the nice sections ~80% good, it makes reviewing long sections of clean data very easy.**

Click "**Apply and continue**"

Review: Edit beats and mark breaks

This is the main section of this script. This is where you will manually go through the deployment and select beats and remove inaccuracies chosen by the automatic detector in the last step.

A window will appear



There is only one setting here, **Min heart rate**. Any beat that follows a gap longer than (60 ÷ Min heart rate) seconds automatically becomes an x (for example, 3 bpm means any gap over 20 s), and this is re-checked after every edit. Set it slightly below the slowest real heart rate you expect, so that genuine slow beats (e.g. during deep dives) are not flagged. **Be careful not to set it too high, even briefly to experiment:** x's created this way are never turned back into dots automatically, and there is no quick way to undo them once the review window has been closed. If it happens while the window is still open, click on the plot and press **Z** to undo the change.

Editing beats

The main controls are listed above the plot, so you don't need to memorize them. Markers show each beat: a **red circle** is a certain beat, a **green circle** is a possible beat (about 80% sure), and a **red x** is a beat with no known beat before it (so no heart rate is calculated for it).

- **Click** on the trace to add a red circle; **Shift+click** adds a green circle; **Ctrl+click** adds a red x. New markers snap to the nearest peak.
- **Click a marker** to remove it.
- **Drag** to move through time, **scroll** to zoom, and use the **left/right arrow keys** to step forward or back.
- **Shift+drag** selects a span (shaded blue); press **Delete** to remove every marker in it (the beat after the span becomes an x), or **Esc** to cancel.

- **Z** undoes the last edit (press repeatedly to undo more).
- **C**, with the pointer over a marker, adds or edits a comment on that beat (commented beats get a black ring).

Preferences

The Preferences section of the panel adjusts how the window behaves for you. These are saved for you on your computer and don't affect the data or other users.

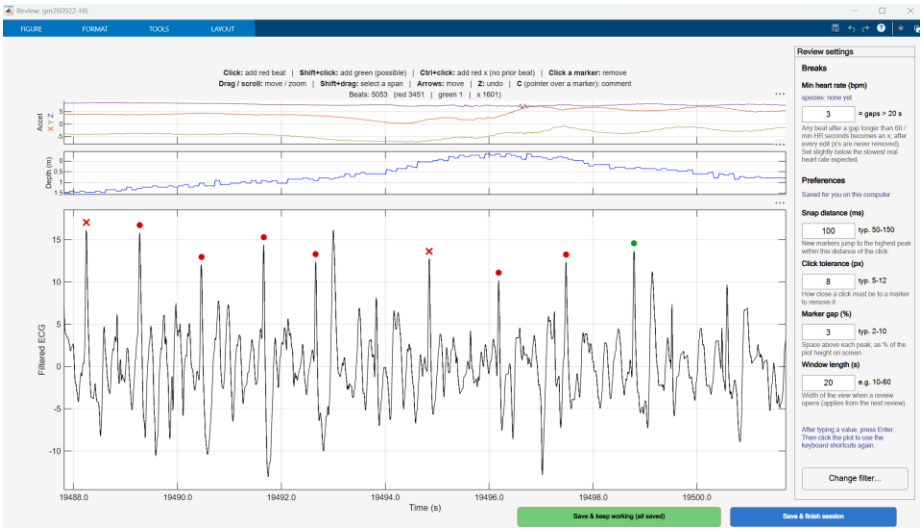
- **Snap distance (ms):** how far from your click the script looks for the highest peak when placing a new marker. Raise it if markers land beside the peak; lower it if they jump to a neighbouring bump.
- **Click tolerance (px):** how close a click must be to a marker to remove it. Raise it if removing markers takes precise aim; lower it if you remove markers by accident.
- **Marker gap (%):** the space between each peak and its marker, as a percentage of the plot height, so markers sit at the same distance at any zoom.
- **Window length (s):** how many seconds are shown when the review window opens (applies from the next review).

After typing a value in the panel, press **Enter**, then click on the plot to use the keyboard shortcuts again.

Change filter… opens the filter window so you can view the data with a different filter. Beats stay where they are; you can then choose to record the new filter in the settings sheets or use it for this session only.

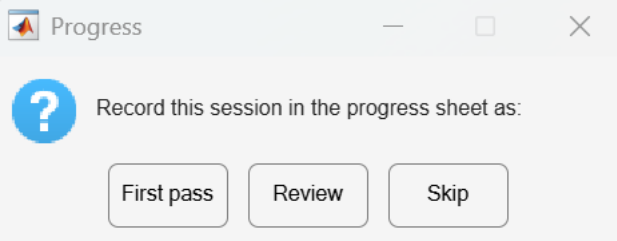
Saving

Save & keep working saves your edits and lets you carry on; it turns orange when there are unsaved changes and green when everything is saved. Save every few minutes, as unsaved edits are lost if MATLAB closes unexpectedly. **Save & finish session** saves, asks for your name, how far you got, and any comments (recorded in the progress sheet), and closes the window. Close the Heartbeats file in Excel before saving, or the save is blocked.



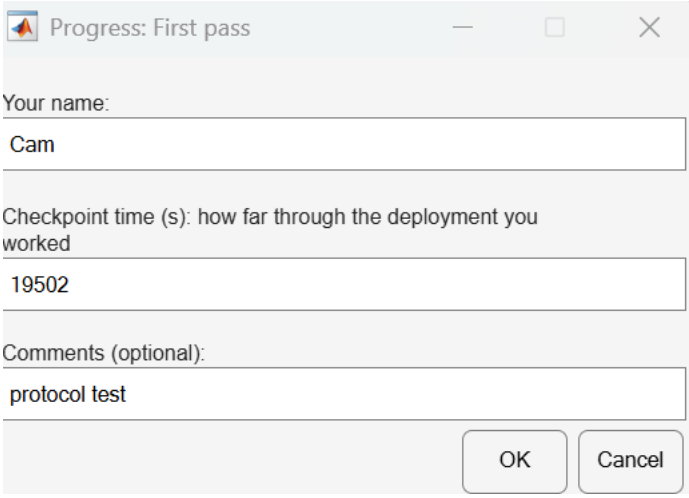
Click "**Save & finish session**" when you are done for the time-being or have completed the deployment.

This popup appears



If you are processing the data for the first time, whether it is your first, second, third, etc. session, click "**First pass**". Only persons reviewing the work of others after the deployment has been finished should click "**Review**".

Enter the requested information. Place the tag off time, or a value close to it, if finished.



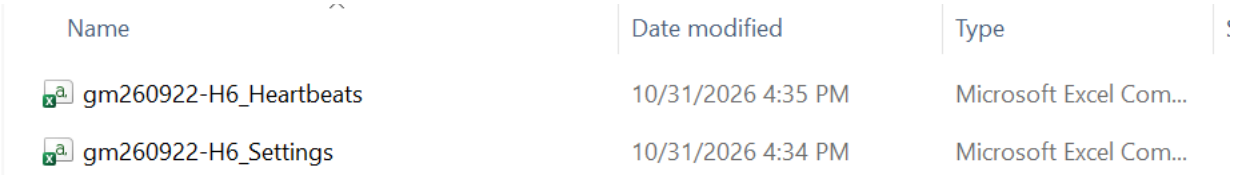
One more popup appears



Click "**Yes**" if complete with the entire first pass. Click "**No**" if more work remains.

Outputs

After finishing, you will have two excel sheets in a processed data folder



Name	Date modified	Type
gm260922-H6_Heartbeats	10/31/2026 4:35 PM	Microsoft Excel Com...
gm260922-H6_Settings	10/31/2026 4:34 PM	Microsoft Excel Com...

<deployment>_Heartbeats looks something like this

	A	B	C	D	E	F	G
1	Time	Certainty	NoPriorBeat	RRInterval	HeartRate	Segment	Comment
2	2680.976	1	1	NaN	NaN	1	
3	2682.16	1	0	1.184	50.67567568	1	
4	2686.004	1	1	NaN	NaN	2	
5	2697.26	1	1	NaN	NaN	3	
6	2700.212	1	0	2.952	20.32520325	3	
7	2703.9	1	1	NaN	NaN	4	
8	2704.652	1	0	0.752	79.78723404	4	
9	2705.3	1	0	0.648	92.59259259	4	
10	2709.88	1	1	NaN	NaN	5	
11	2710.52	1	0	0.64	93.75	5	
12	2711.16	1	0	0.64	93.75	5	

Time: the time of the R-wave peak for that beat, in seconds from the start of the recording (the same time base as the raw data file, not from tag-on). It is the highest point of the filtered ECG at that beat, to the nearest sample.

Certainty: “1” for all red circles, “2” for any green circles. This allows the user to select beats they are not 100% confident in, whenever that may be necessary, and for any supervisors to easily review that section and/or filter out those data.

NoPriorBeat: The first heartbeats within a segment. There was likely noise before this heartbeat, thus this particular beat does not have a R-R interval or instantaneous heart rate (both columns appear as NaN).

RRInterval: The duration (in seconds) between the previous heartbeat and the current heartbeat.

HeartRate: Instantaneous heart rate of that beat, calculated as 60 divided by the R-R interval

Segment: Segment: a number (1, 2, 3, …) identifying each continuous stretch of beats, meaning a run of beats with no known gaps between them. A new segment starts at every beat with

NoPriorBeat = 1 (a red x), and continues until the beat before the next x. The first beat of the deployment always starts Segment 1.

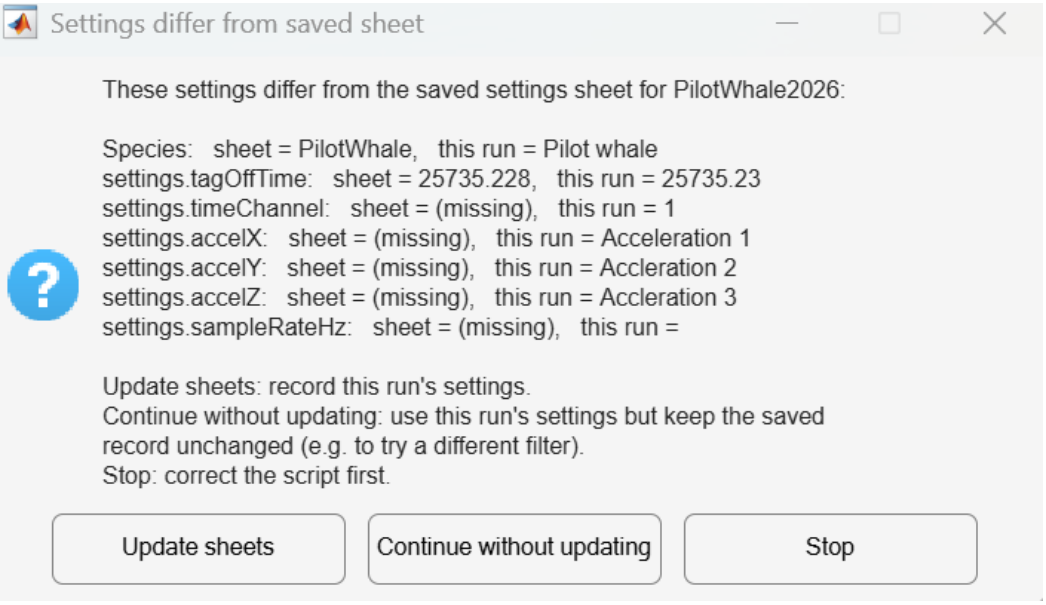
Your deployment_Settings excel sheet will have columns that look something like this

M	N	O	P	Q	R	S	
settings.highPassHz	settings.lowPassHz	settings.filterOrder	settings.invertECG	settings.minStandOut	settings.minSteepness	settings.maxWidthMs	se
1	40	4	TRUE	4	4	80	

*This is not all of them shown above

This is used by the script when returning to work on or review deployments.

If you rerun the script with different settings, you will get a popup box that look like this, explaining what is different. You have three options, to update the Settings sheet, to continue without updating the settings sheet, or to stop the run.



There are also three new excel sheets that have been added to your upper-level project folder



HeartRate_Progress (or whatever name of folder you chose instead of HeartRate)

Deployment	Processed	ProcessedDate	CheckpointTime	FirstPassComplete	ReviewedBy	ReviewDate	ReviewCheckpointTime	ReviewComplete	Comments
20-Oct-2026-Frankie-2-suction-cups-	Cam	10/31/2026	4127	Yes					2026-10-28 Cam (First pass): Testing comment function.;
21-Oct-2026-Frankie-2-suction-cups-	Cam	10/31/2026	finished	Yes					2026-10-30 Cam (First pass): Great deployment!; 2026-1(
26-Oct-2026-Nando-2 ecg suction cups	Cam	10/31/2026	20008	Yes					2026-10-29 Cam (First pass): Finished. Really messy depl
27-Oct-2026-Chelo-2ecg suction cups-1ecgsuctioncup	Cam	10/31/2026	16733	Yes	Andreas		16875	Yes	2026-10-28 Andreas (Review): Cam/Andreas review; 2026
6-Oct-2026-Nando-1suctioncup	Cam	10/31/2026	23262	Yes					2026-10-16 Cam (First pass): None; 2026-10-31 Cam (Fir
gm260922-H6	Cam	10/31/2026	19502	No					2026-10-31 Cam (First pass): protocol test

This is a running list of your progress across all of your deployments. At this stage, prior to any review, only columns A-E should be populated. These come from those last popup boxes after the manual processing of the data.

HeartRate_Reference_Settings

A running list of the settings used for every deployment, one row per deployment, sorted by species. It is updated automatically whenever a deployment's settings are recorded, so it never needs editing by hand.

It serves two purposes. For people, it shows what has worked before: someone new to a species can see the filter, detection and break settings used for earlier deployments of that species. For the script, it provides the **starting values** for a new deployment (taken from the most recently processed deployment of the same species) and the **species ranges** shown in blue next to each setting in the tuning windows and review panel.

Species	Deployment	settings.rawFile	settings.tagOnTime	settings.tagOffTime	igs.timeCh	igs.ecgCh	igs.depthCl	ttings.acc	ttings.acc	ttings.acc	gs.sample	ngs.highPa	ngs.lowPa	ings.filter	Oings.inver	tigs.minSta	gs.minSte
bb260613-A2	bb260613-A2	bb260613-A2.txt	507.74	4189.704	1	2						1	40	4	FALSE	20	3
gm240124-C1	gm240124-C1	gm240124-C1.txt	1100	17000	1	ECG	Depth (m)					1	40	4	FALSE	20	3
gm240228-H1	gm240228-H1	gm240228-H1.txt	0.184	23261.64		ecg	depth					1	40	4	TRUE	20	3
mn250303-H4	mn250303-H4	mn250303-H4.txt	2679.44	24973.16	1	ECG	Depth (m)	celeration	celeration	celeration	3	1	40	4	TRUE	4	4
mn260228-J3	mn260228-J3	mn260228-J3.txt	2679.668	25735.23	1	ECG	Depth (m)	celeration	celeration	celeration	3	1	40	4	TRUE	4	4

HeartRate_Deployment_Information

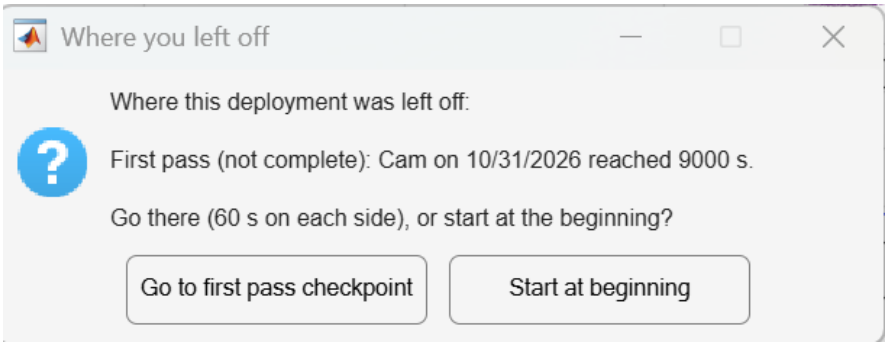
Deployment	Species	Complete	TagOnTi	TagOffTim	DeploymentDuration_h	NumHeartbeats	NumCerta	NumPossi	NumSegm	NumRRInt	MeanSegn	MedianSe	SDSegmer	MeanBeat	MedianBe	SDBeatsPe	PercentClean	ECG	MeanHear	MedianHe	SDHea
20-Oct-2026-Frankie-2-su	Loggerhead	First pass complete	507.74	4189.7	1.023	316	312	4	20	296	1.82	0.73	2.96	15.8	9	20.24	59.32	9.69	7.81	5	
21-Oct-2026-Frankie-2-su	Loggerhead	First pass complete	550	15600	4.181	1675	1651	24	101	1574	1.78	0.85	2.83	16.58	9	19.25	71.86	10.49	8.52	5	
26-Oct-2026-Nando-2 ecg	Loggerhead	First pass complete	1200	20355	5.321	122	96	26	12	110	1.91	0.48	4.41	10.17	5	16.79	7.19	7.39	4.13	7	
27-Oct-2026-Chelo-2ecg s	Loggerhead	Review complete	1100	17000	4.417	1111	1111	0	83	1028	1.73	0.69	4.21	13.39	5	26.86	54.25	8.64	6.71	4	
6-Oct-2026-Nando-1sucti	Loggerhead	First pass complete	0.184	23261.6	6.462	420	356	64	30	390	3.18	2.47	3.76	14	12	12.85	24.59	5.86	3.81	8	
gm260922-H6	Pilot whale	In progress	2679.2	25045.8	6.213	5053	5052	1	1601	3452	0.08	0.05	0.1	3.16	2	2.44	35.07	35.29	34.01	16	

A summary of every deployment, one row per deployment, sorted alphabetically. It is refreshed automatically every time a review is saved, so it always reflects the latest beats. Use it to compare deployments, track progress, and judge data quality at a glance.

- **Deployment details:** deployment name, species, status (In progress / First pass complete / Review complete, taken from the progress sheet), tag on and off times (s), and deployment duration (h).
- **Beat counts:** total heartbeats, certain beats, possible beats, and the number of usable RR intervals (beats with a known beat before them).
- **Segments:** a segment is an unbroken stretch of beats between x's. The sheet gives the number of segments, their mean, median and SD length (min), and the mean, median and SD number of beats per segment.
- **Data quality:** PercentCleanECG, the percentage of the tag-on period covered by segments, i.e. how much of the deployment has a continuous, usable heart rate record.
- **Heart rate:** mean, median and SD heart rate (bpm), from the usable RR intervals.
- **LastUpdated** (date of the last save) and **Comments** (copied from the progress sheet).

Returning to complete a first pass (continuing where you left off)
Open the script and click "**Run**"

Choose "**Existing deployment**"
Select the deployment folder
"**Continue review**"
When returning, you will get this popup that allows you to jump to where the last user left off.

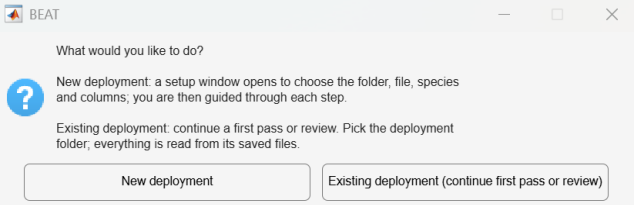


Resume editing!

Using the script to review and modify processed data
For anyone supervising the work of others or checking your own work

Returning to review work

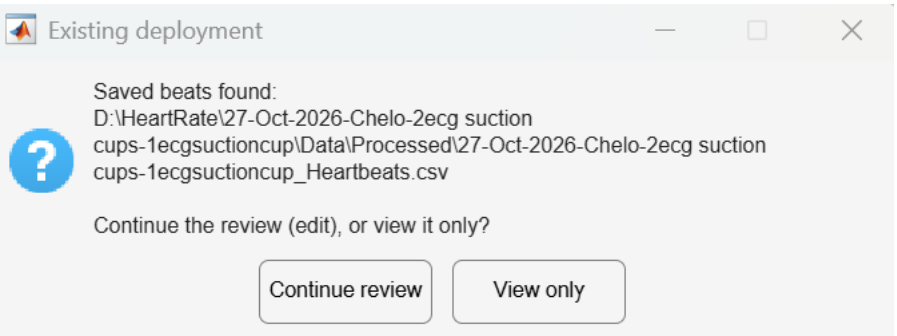
Open the script and click "**Run**"



Choose "**Existing deployment**"

Select the deployment folder

You will get a popup box that looks like this



Continue review: reopens the saved review from "deployment_Heartbeats", with all beats, x's and comments exactly as they were saved, and all editing options available. The window opens at the start of the tag-on period; use the checkpoint time in "HeartRate_Progress" to find where work stopped.

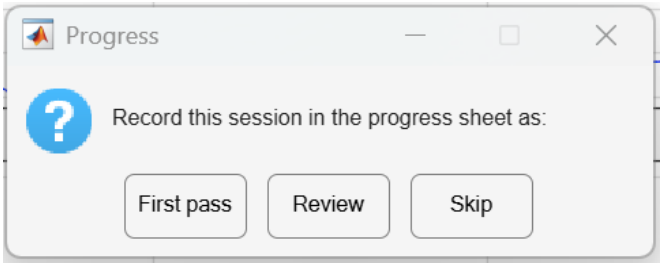
View only: shows the saved review with editing switched off, so nothing can be changed by accident. Useful for checking someone else's work. Close the window with the **Close** button.

If you have already started to review the data but stopped prior to finishing, you should receive a pop up that looks like this.

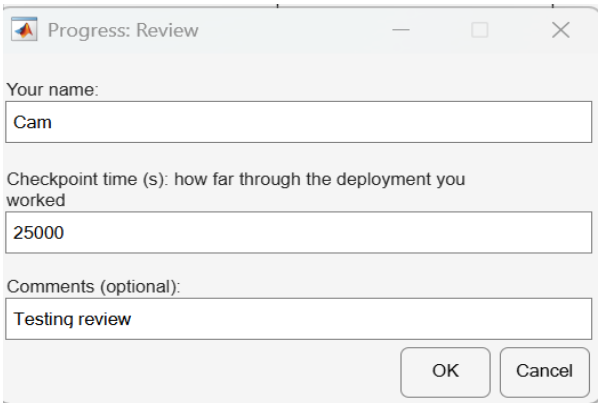


Be sure to save as you review and to click "Save & finish session" when complete!

A popup box will appear, click "**Review**"

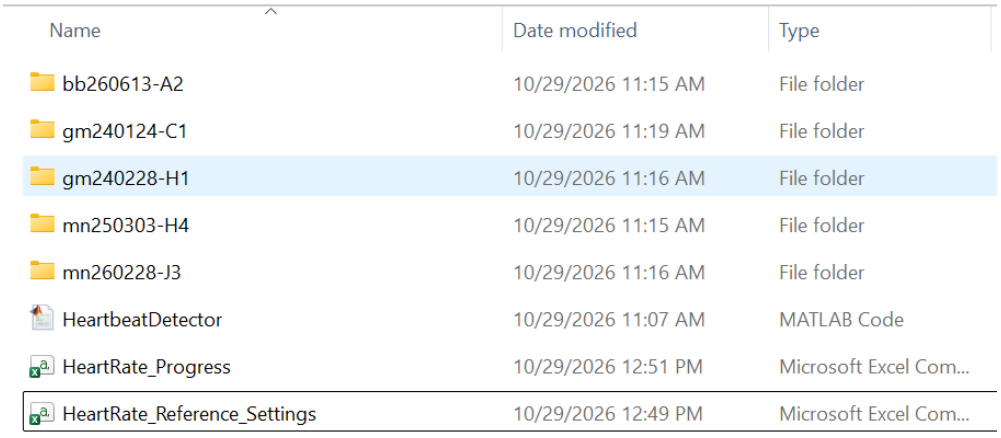


The next popup box will appear. Enter your name, the time in the deployment that you have reviewed through, and any comments you want to make.



And one final box will appear. Click **"Yes"** if you have completed the deployment review and **"No"** if not.

At this stage, your "HeartRate_Progress" sheet should have the review information entered automatically. This will allow you to see which deployments have been completely reviewed.



Name	Date modified	Type
bb260613-A2	10/29/2026 11:15 AM	File folder
gm240124-C1	10/29/2026 11:19 AM	File folder
gm240228-H1	10/29/2026 11:16 AM	File folder
mn250303-H4	10/29/2026 11:15 AM	File folder
mn260228-J3	10/29/2026 11:16 AM	File folder
HeartbeatDetector	10/29/2026 11:07 AM	MATLAB Code
HeartRate_Progress	10/29/2026 12:51 PM	Microsoft Excel Com...
HeartRate_Reference_Settings	10/29/2026 12:49 PM	Microsoft Excel Com...